# Zero-Knowledge Academic Verification (ZKAV) Protocol: The First Cross-Paradigm Non-Disclosure Academic Verification Framework Pioneered by the SGT Theory

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## Abstract

The Spatial Pressure Gravitational Theory (SGT), an original gravitational and cosmological framework constructed from first principles, encounters a structural dilemma when interfacing with mainstream academic evaluation systems: the inherent conflict between intellectual property (IP) confidentiality and the verification of academic validity. Traditional peer review often demands full disclosure of internal derivations, posing a severe risk of misappropriation for unpublished original work. To resolve this, the author pioneered the Zero-Knowledge Academic Verification (ZKAV) protocol, grounded in SGT research and development, and globally launched it for free academic use. This protocol employs a standardized closed-loop task workflow. Under the strict constraint of concealing SGT's core field equations, intermediate calculations, and unpublished parameters, it allows domestic and international research institutions to propose any gravitational or cosmological physical propositions. The author independently performs all deductions based on the SGT framework **via a self-developed SGT solver**, outputting only observable results that can be independently recalculated by the collaborators using their own theoretical systems. Objective judgments on SGT's validity and model discrepancies are then achieved through numerical comparison. ZKAV utilizes third-party email services to establish a fully auditable communication chain with judicial evidentiary validity. Furthermore, the protocol is not bound to a single email provider; any compliant email platform capable of retaining immutable communication records can be adapted for implementation. This paper systematically elucidates the creation logic, standardized execution workflow, and forensic evidence design of ZKAV, clarifying that SGT is currently the world's first theoretical system to fully implement and routinely open ZKAV verification tasks. We sincerely invite astrophysical and theoretical cosmological research institutions worldwide to conduct zero-leakage, traceable academic exchanges and joint verifications via this protocol.

**Keywords:** SGT; Spatial Pressure Gravitational Theory; Zero-Knowledge Academic Verification; Cross-paradigm verification; Electronic evidence

retention; Open academic protocol

## 1. Introduction

The Spatial Pressure Gravitational Theory (SGT) is independently constructed from first principles by the author. It interprets the essence of gravity through the tension effect of a vacuum elastic spatial-energy medium, theoretically eliminating the singularities in cosmology and black holes, and deriving a series of quantitative, observable predictions that distinguish it from General Relativity, such as an ISCO radius of 4.23M and a Hawking radiation temperature coefficient of 0.866. During the phase of seeking peer academic verification, the inherent review model of the industry forms an intractable "trust-disclosure" dilemma: without the complete theoretical derivation text, academic institutions are reluctant to invest research resources in carefully evaluating a new theory; conversely, if the original author fully discloses the core content, they face the realistic risks of having their key academic achievements misappropriated without compensation or subjected to unconstructive, fragmented nitpicking detached from the holistic framework.

Addressing these practical challenges, the author draws upon the underlying philosophical concepts and methodological paradigm of zero-knowledge interactive proofs in cryptography (note: this protocol is an academic verification paradigm, not a foundational cryptographic algorithm) to originally design the Zero-Knowledge Academic Verification (ZKAV) protocol. Relying on the SGT framework, the author has become the world's first researcher to implement and routinely open this verification mechanism. ZKAV breaks away from the two established academic verification modes—full-text submission and full-source-code open-source reproduction—establishing a new interactive paradigm of "initiator-proposed propositions, creator's closed-book solving, and cross-verification based on results." This paper serves as the formal academic documentation of the ZKAV protocol and defines the academic positioning of SGT as the original creator and the first practical carrier of this verification paradigm.

## 2. The SGT-Original ZKAV Protocol: Conceptual Definition and Standardized Operational Workflow

### 2.1 Three Core Principles of the Protocol

All rules of ZKAV are independently formulated by the author, and its implementation strictly adheres to three core rigid principles:

1. **Zero-Knowledge Confidentiality Principle:** The SGT side will never disclose internal field equations, pending parameters, or step-by-step derivation processes externally, delivering only the final calculation results.
2. **Task Equality Principle:** The verification propositions are entirely formulated by the initiating research institution, the calculation is independently completed by the SGT side, and the right to verify the

results belongs to the initiator. Rights and responsibilities are mutually equal.

3. **Evidentiary Preservation Principle:** All task communications are conducted through third-party compliant email services with timestamp preservation capabilities. The native communication records automatically form a complete chain of evidence that is notarizable and legally admissible in court.

### 2.2 Definition of the Three Parties in the Protocol

- **Initiator (P):** Research entities intending to verify the SGT theory, including university physics departments, observatories, research institutes, and academic journal editorial boards.
- **Prover (S):** The creator of the SGT theory, who is simultaneously the designer and the first implementing entity of the ZKAV protocol.
- **Witness (T):** A third-party email service provider with compliant data retention qualifications. The protocol does not restrict the designated platform; the demonstration contact email is zhijundi@qq.com.

### 2.3 Six-Step Closed-Loop Standardized Workflow

| Step | Executor | Specific Action | Scope of Information Disclosure |
|---|---|---|---|
| **1. Task Initiation** | P | Sends research topics, boundary parameters, and | No theoretical information |

| | | | |
|---|---|---|---|
| | | expected output metrics via standard email template. | disclosed. |
| **2. Task Confirmation** | S | Replies to acknowledge receipt and agrees on the delivery timeline. | No theoretical information disclosed. |
| **3. Independent Calculation** | S | Completes the closed-book solution **using the proprietary SGT solver** based on the SGT theory; all calculation materials and **solver execution logs** are retained and archived internally. | Zero internal information disclosed. |
| **4. Result Delivery** | S | Sends only the final numerical solutions, spectral functions, or other observable conclusions. | Only objective output results. |
| **5. Independent Verification** | P | Recalculates and compares data discrepancies using its own mature theoretical system. | No theoretical information disclosed. |
| **6. Verification Receipt** | P | Voluntarily issues a conclusion receipt, noting data consistency and model deviations. | No theoretical information disclosed. |

**Hard Constraint of the Protocol:** SGT has no obligation to explain the principles behind the derivation of the results. Initiators who do not accept this constraint may waive their verification application.

### 2.4 Objective Background of SGT's Pioneering Implementation

Prior to the advent of ZKAV, no similar task-based, non-disclosure academic verification standard existed in the field of fundamental physics domestically or internationally. This paradigm was innovatively created by the author specifically to break through the review barriers hindering the implementation of SGT. From conceptual conception and workflow design to web specification publication (sgt.ac.cn/zkav.html), all R&D work was completed independently by the author and officially released in 2026. SGT is not only the sole original source of the ZKAV protocol but also currently the only physical theory system globally that routinely opens itself to ZKAV verification tasks.

## 3. Chain of Evidence Architecture: Jurisprudential and Technical Support of Third-Party Email

The selection of zhijundi@qq.com as the demonstration contact email is based on a comprehensive consideration of electronic evidence standardization regulations: First, the email operator is an independent third-party enterprise whose server logs are not controlled by any party to the protocol, meeting the requirements of neutral evidence retention carriers in electronic evidence regulations. Second, the system automatically generates immutable sending and receiving timestamps, and the original email packets are retained in the cloud long-term. Third, judicial and notary organs can legally request original communication records from the service provider, possessing a complete judicial forensic pathway. The specific admissibility standards and retrieval procedures for electronic evidence shall be subject to the relevant electronic evidence regulations of the jurisdiction where the initiator is located.

Simultaneously, the protocol text explicitly states: ZKAV rules have no service provider binding restrictions. Any compliant email (commercial emails, research institute institutional emails, etc.) that can retain complete sending/receiving logs and traceable timestamps can serve as the protocol's communication carrier. When other researchers reuse ZKAV, they are free to change the contact email, needing only to cite the original source of the protocol in their results. During the protocol design phase, the author deliberately avoided using a self-hosted email server, eliminating potential disputes over evidentiary validity at the architectural level.

## 4. Discussion: Academic Paradigm Innovation Brought by the Pioneering of ZKAV

### 4.1 Horizontal Comparison with Traditional Academic Verification Models

| Comparison Dimension | Traditional Peer Review | Full Open-Source Reproduction | ZKAV (SGT Pioneered) |
|---|---|---|---|
| **Degree of Derivation Disclosure** | Partially public | All derivations and source code public | Zero internal derivation disclosure throughout |
| **IP Risk** | Moderate | Extremely High | Extremely Low |
| **Participation** | Requires | Requires setting up | Simply send a |

| | | | |
|---|---|---|---|
| **Threshold** | reading the entire theoretical manuscript | the corresponding numerical computing environment | proposition via email |
| **Legal Validity of Evidence** | Weak | Moderate | Judicial-grade evidentiary validity |
| **Right to Choose Propositions** | Fixed topics by reviewers | Preset examples by the original author | Custom topics by the verifier |
| **Implementation Instances** | Widely used in academia | Adopted by some open-source projects | World's first implemented and running instance (SGT) |

ZKAV is not intended to replace the traditional peer review system but to fill the "pre-trust" verification gap between the original formation of a new theory and its formal submission. For all future academic practices conducted using the ZKAV standard, SGT and the author of this paper will be recognized as the original creators of this paradigm.

### 4.2 Multidimensional Academic Benefits of SGT Implementing ZKAV

First, it secures the right to define verification rules, shifting from passively waiting for expert review to proactively providing a standardized verification channel. Second, it comprehensively protects SGT's unpublished core content, avoiding the risk of original work leaking. **Furthermore, the underlying SGT solver ensures that the "black-box" calculations possess software-engineering-level determinism and stability,**

**significantly enhancing the reliability of the verification results.** Third, every round of email task exchanges is automatically archived, continuously accumulating auditable, formal academic communication evidence.

### 4.3 Universal Expansion Potential of the Protocol

The full text of the ZKAV protocol is open without copyright fees or authorization thresholds. Original theories in any direction—gravitational theory, quantum field theory, particle physics, etc.—can directly reuse or fine-tune it to adapt to their own research, assisting the entire fundamental physics field in perfecting the verification system for nascent theories.

## 5. Conclusion

This paper formally declares: The Zero-Knowledge Academic Verification (ZKAV) protocol was independently originated and developed by Zhijun Li. In 2026, it was pioneered and implemented relying on the SGT system and opened free of charge to the global academic community. SGT thus becomes the world's first gravitational theory to accept unlimited verification tasks on any external physical proposition without leaking its internal derivation system. The protocol rules are neutral, open, and not bound to any specific email service or user entity. We sincerely invite research institutes worldwide to conduct academic exchanges via the ZKAV channel.

**ZKAV Demonstration Task Contact Email:** zhijundi@qq.com

**Complete Protocol Specifications and Bilingual Email Templates:**

https://sgt.ac.cn/zkav.html

**SGT Solver (Computational Engine for ZKAV):** https://sgt.ac.cn/solver/

## Protocol License Declaration



**Supplementary Note on Email Usage:** The demonstration email is merely an implementation instance; the protocol itself does not restrict the communication carrier. Any email system that meets compliant evidence retention requirements can deploy ZKAV.

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## Conflict of Interest Statement

The author of this paper is the sole original designer and implementing practitioner of the ZKAV protocol. The protocol development has no external funding injection from project grants or corporate sponsorships, and there are no potential conflicts of interest related to this work.